

Effects of the toxic metal zinc on the growth, development, and reproduction of the wolf spider *Pardosa laura* through its food *Drosophila melanogaster*

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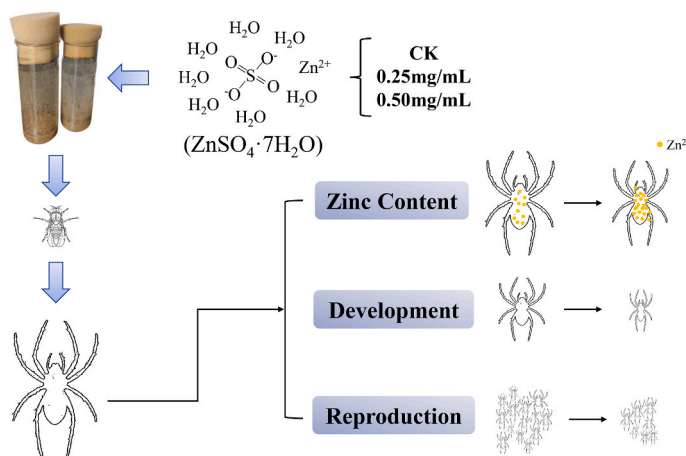
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HIGHLIGHTS

- Toxic metal Zn could accumulate in the spider *Pardosa laura* through its food *Drosophila melanogaster*.
- Different levels of toxic metal contamination can cause serious harm along the food chain.
- Zn had a significant inhibitory effect on the growth, development and reproduction of *P. laura*.
- This study will provide a new perspective for evaluating the Zn metal effects on spiders.

GRAPHICAL ABSTRACT



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ABSTRACT

In recent years, toxic metal pollution has become a serious problem, and its influence on human society and the ecological environment has become a hot topic for people. Toxic metal pollution is ubiquitous in the environment, and it can affect the growth, development, and reproduction of organisms through food chain transmission. In this study, we used ZnSO_4 and set three different Zn concentrations [0 mg/mL (CK), 0.25 mg/mL, and 0.50 mg/mL] to study the enrichment of toxic zinc metal in the wolf spider *Pardosa laura* through the food chain (medium-*Drosophila melanogaster*-*P. laura*) and the effects on the growth, development, and reproduction of *D. melanogaster* and *P. laura*. The results of the study showed that the Zn content in *D. melanogaster* and *P. laura* significantly increased with the increase of Zn concentration in the culture medium, reflecting the obvious food

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chain enrichment phenomenon. The inhibitory effect of zinc treatment on the growth, development, and reproduction of *D. melanogaster* was significant and more pronounced with increasing Zn concentration. The transfer of zinc through the food chain also had a significant inhibitory effect on the growth, development, and reproduction of *P. laura*. The developmental duration and preoviposition period were significantly prolonged. Moreover, the survival rate, body size, and egg laying amount were significantly reduced. This study will enrich and improve the research on the effects of toxic Zn metal pollution on spiders and provide a theoretical basis for monitoring and evaluating the environmental quality using farmland spiders.

1. Introduction

Toxic metal pollution has been a hot issue of global concern for environmental pollution. China has the second largest zinc reserves in the world (Liu et al., 2017), and with the development of modernisation, zinc resources are widely used in industrial and agricultural production, such as metal smelting, drug manufacturing, and the food industry, making zinc pollution the most common kind of toxic metal pollution in the environment. However, in most cases in China, soil background Zn levels have been well below environmental quality standard limits, and its cumulative characterisation and risk analysis have not received the same attention as that given to toxic metals such as Cd, Pb, As, Hg, and Cr (Li et al., 2023; Yuan et al., 2020). A recent study (Zhang et al., 2022) showed that the contents of seven toxic metals, including Zn, exceeded the limits to varying degrees in the soils of agricultural land in the industrial areas of the Yangtze River Economic Zone in China (evaluated by soil environmental quality risk control standard for soil contamination of agricultural land, GB 15618-2018), with Zn second only to Cd. As a result, global research on the environmental and biological effects of zinc pollution is increasing.

Toxic metals can invade animals directly through breathing, skin, and ingestion and can also enter their bodies through the food chain, causing certain positive and negative effects on their growth and development, among other processes (Green and Walmsley, 2013). The enrichment of a toxic metal in animals can be determined by the enrichment factor (Zhang et al., 2007), which is a common indicator of toxic metal contamination, with a higher factor indicating a greater likelihood of toxic effects on animals. Some studies have shown that the variability of enrichment factors is influenced by species differences and different toxic metal elements (Ye and Li, 2011), which depends on the ability and sensitivity of the species to accumulate metal contaminants (Sebbio et al., 2014).

Some studies have found significant gender and tissue differences in arthropod toxic metal enrichment (Ardestani et al., 2014; Kalman et al., 2015; Luoma and Rainbow, 2005; Rainbow and Luoma, 2011; Stankovic et al., 2013). In addition, biomagnification has been found to occur by toxic metals being transferred through the food chain (Hendrickx et al., 2003), and based on their structural characteristics, biological disposition, and bioactive potential, these substances may be highly persistent in the environment (Noyes and Lema, 2015). However, there are also findings that indicate that toxic metals do not bio-magnify in the beet aphid through food chain action (Crawford et al., 1990). Instead, an increase in toxic metal concentrations over generations occurred, reflecting the relationship between toxic metal accumulation in the insect and the concentration of toxic metals in its food and the duration of the stress (Bromenshenk et al., 1991; Heliövaara and Väisänen, 1989; Ruohomäki et al., 1996). The toxicity of toxic metals in animals can affect growth, reproduction, sex ratios, and courtship behaviour and is complicated by the concentration of toxic metals, the distance from the source, the site of action of the toxic metals, and the developmental stage of the animal (Dai, 2004; Du Laing et al., 2002; Lukkari et al., 2004; Reinecke et al., 2001; Salminen and Haimi, 2001; Skubala and Kafel, 2004). Current research on Zn contamination is mainly on the environment (water, soil, and atmosphere), plants, and animals (poultry species as well as insects), but relatively little research has been conducted on spiders.

Pardosa laura is widely distributed in China and is one of the dominant spiders in farmlands, being an important natural enemy in many agricultural fields, such as rice, vegetables, and fruit trees, and playing a positive role in farmland pest control (Wang and Yan, 2000). As one of the important natural enemies of farmland pests, its population size is closely related to the effectiveness of its biological control of pests. Toxic metal pollution not only has a negative impact on the ecosystem but also has a negative impact on biodiversity. Organisms can take up and enrich toxic metals from the environment in their bodies by means of food chain transfer (Green and Walmsley, 2013), which can have certain effects on their growth, development, and physiological functions (Grześ, 2010; van Ooik and Rantala, 2010). Different metal ions are stabilised differently in plants, and the less stable Zn^{2+} ions are more likely to be transported suddenly from the soil through the plant to its above-ground parts, which in turn affects above-ground arthropods (Zhu et al., 2007). Numerous studies have reported the effects of Zn on plants, insects, aquatic animals, and ruminants (Ye and Li, 2011), but fewer studies have been conducted on spiders. Due to polyphagy and easy nutrient uptake, carnivorous spiders bioaccumulate the highest concentrations of toxic metals compared to other invertebrates (Hendrickx et al., 2003; Li et al., 2016a, 2016b; Lv et al., 2021; Tang et al., 2022; Wang et al., 2019, 2021). Therefore, there is a need to investigate the effects of different Zn concentrations on enrichment in *P. laura*. In this study, the effect of Zn on *P. laura* was investigated by means of food chain transfer. This study will enrich and improve the study of the effects of zinc pollution on spiders and provide a theoretical basis for the use of farmland spiders for monitoring, evaluation, and pollution prediction of farmland environmental quality.

2. Materials and methods

2.1. Test fruit flies and rearing

D. melanogaster were cultured in this laboratory for multiple consecutive generations. *D. melanogaster* were reared in culture tubes (12 cm height × 5 cm diameter) in an artificial climate chamber at a temperature of 25 °C, relative humidity of 35%–45%, and a photoperiod of 14L:10D. The recipe for the *D. melanogaster* medium (Wen et al., 2020) was as follows: 240 mL tap water + 22 g maize flour + 4 g yeast + 22 g sucrose + 1.6 g agar powder + 1 mL propionic acid + 0.1 g benzoic acid. Zinc sulphate heptahydrate ($ZnSO_4 \cdot 7H_2O$) was also added to the *D. melanogaster* medium. $ZnSO_4 \cdot 7H_2O$ has been applied to crop zinc fertilisation (Shivay et al., 2015) and other entomophagous physiological studies (Ahmad et al., 2021). Based on the degree of soil contamination in China (GB 36600-2018, China; GB 15618-2018, China), three fruit fly mediums containing different Zn^{2+} concentrations of CK (control sample, 0 mg/mL), 0.25 mg/mL, and 0.50 mg/mL were prepared, and the adult fruit flies obtained were used for the determination of Zn content.

2.2. Collection and rearing of spiders

The female spiders with egg sacs were collected from the Hubei University campus (30°34'N; 114°19'E) and reared individually in glass tubes (6 cm height × 2 cm diameter) with water-soaked sponges placed at the bottom of the tubes. The mouth of the tube was closed with a

skimmed cotton ball. Spiders were fed with adult *D. melanogaster* (Zn-containing or not) twice weekly, which could satisfy their nutrition requirements. The spiderlings (2nd instar) hatched from the egg sacs were divided into three groups and each group consist of 20 individuals. Spiders were kept under lighting (12L:12D), temperature (L: 25 °C, D: 18 °C), and humidity conditions (70% ± 10%). The adult *P. laura* spiders obtained were used to determine zinc content.

2.3. Experimental design for Zn enrichment in *D. melanogaster* and *P. laura* through the food chain

Ten adult *D. melanogaster* or two adult *P. laura* were randomly selected as one sample for the different Zn concentration treatments, with five biological replicates for each treatment. *D. melanogaster* or *P. laura* were detected with their whole bodies. The samples were dried at 60 °C for 48 h. Several counts were taken and weighed, then placed in a microwave digestion tube for digestion. The digest was driven to acid and fixed to 50 mL with deionised water, and the supernatant was added to an inductively coupled plasma emission spectrometer (ICP-OES) device (PERKINELMER Optima 8000, USA) for determination. Without weighing the sample, the blank sample was ablated for determination as described above. The zinc contents in *D. melanogaster* and *P. laura* were obtained by calculating the measured values in the measurement and blank groups.

The formula for calculating zinc content was:

$$\rho = (\rho_1 - \rho_0) \times f \times V_1 \div (m \times \omega_{dm} \div 100)$$

where ρ was the sample result (mg/kg), ρ_1 was the measured value (mg/L), ρ_0 was the mean blank value (mg/L), f was the dilution factor, V_1 was the volume of the sample in the volumetric flask, m was the sample mass (g), and ω_{dm} was the dry matter content (%).

2.4. Determination of biological characteristics of *D. melanogaster* under different zinc treatments

Five 5-day-old *D. melanogaster* adults (female:male = 3:2) were incubated in the laboratory at room temperature, placed in three different Zn concentrations of *D. melanogaster* medium, plugged with sponge plugs, and placed in the artificial climate chamber (temperature: 25 °C, relative humidity 30% ± 5%, lighting 14L:10D). Six hours later, the adults were released and the number of eggs laid by each *D. melanogaster* was recorded using a magnifying glass. *D. melanogaster* cultures with eggs were returned to the artificial climate chamber under the same conditions. After 24 h, *D. melanogaster* eggs were observed for hatching, and hatching rates were calculated separately. The number of pupae, the number of adults, and the time from egg hatching to adults were observed and recorded daily for each treatment. Thirty biological replicates of each treatment were used.

Freshly emerged (unmated) *D. melanogaster* (female:male = 1:1) were randomly selected from each treatment and killed with 75% alcohol, then filter paper was used to dry the alcohol, and the weight of each fly was measured on an electronic scale (Model FA1004 N; Shanghai Precision Instruments). The body length of the flies was measured under a microscope (Model DFC495; Leica, Germany) using an eyepiece micrometre. As there were inherent differences in body size between male and female *D. melanogaster*, we compared the effect of toxic metal zinc on body weight and body length of females and males separately. Thirty biological replicates of each treatment were used.

To determine the longevity of adult *D. melanogaster*, newly emerged fruit flies were placed individually in new mediums with three Zn concentrations. The surviving days of the *D. melanogaster* under different Zn concentration treatments was recorded. Thirty biological replicates of each treatment were used.

2.5. Determination of *P. laura* growth, development, and reproduction through the food chain

P. laura were fed *D. melanogaster* reared in mediums with different Zn concentrations (0, 0.25, and 0.50 mg/mL) from the age of three days, three times per week, and the development conditions were observed and recorded daily. Treatments with 20 spiderlings were performed with three replicates of each treatment.

After the *P. laura* matured, the body weight (mg) of the spiders was measured using an electronic scale (FA1004 N; Shanghai Precision Instruments), and the body length (mm) of the spiders was measured under a microscope (DFC495; Leica, Germany) using an eyepiece micrometre. For each treatment, there were three replicates, each with 20 individuals.

P. laura were fed with *D. melanogaster* (reared on three different Zn concentration mediums) until maturity. Male and female spiders were allowed to mate. Heterosexual cannibalism during mating was prevented by providing the spiders with adequate food and water. After mating, the female and male spiders were returned to their original glass finger tubes (6 cm height × 2 cm diameter). The spiders were fed every three days, water was added to the glass finger tubes to maintain the humidity of the sponge inside, and the tubes were cleaned regularly. The female spiders were observed every day for egg laying, and the number of egg sacs laid were recorded daily. When the egg sacs were mature, they were gently swept off the females' spinnerets with a brush. The female spiders' egg laying rate, the first egg sac pre-laying period (mating to egg laying time), and the number of eggs laid in the first egg sac were counted. Fifteen mating experiments were conducted for each treatment.

2.6. Statistics

The data are expressed as mean ± standard error. Differences in zinc content, egg laying, hatching rate, pupation rate, fecundity and developmental duration, body weight, and body length in *D. melanogaster* and *P. laura* were analysed by one-way analysis of variance (ANOVA), and multiple comparisons of means were analysed by the least significant difference (LSD) method. Differences in the survival of *D. melanogaster* were analysed using a log-rank test; differences in survival of *P. laura* and egg laying rate of female *P. laura* were analysed using chi-squared tests. Statistical tests were performed using the IBM SPSS Statistics 26 program for Windows (SPSS Inc., Chicago, USA), and GraphPad Prism (8.0) was used for graphing.

3. Results

3.1. Enrichment of zinc in *D. melanogaster* and *P. laura* through the food chain

As shown in Fig. 1a, the Zn content in *D. melanogaster* increased significantly with increasing Zn concentration in the medium (LSD test, $P < 0.001$). The Zn content in *D. melanogaster* reared in Zn medium concentrations of 0.25 mg/mL and 0.50 mg/mL were 3.76 and 6.56 times higher than in the CK group, respectively. There was a significant difference in the Zn content in *P. laura* (Fig. 1b; LSD test, $P < 0.001$). The Zn concentrations of the 0.25 mg/mL and 0.50 mg/mL treated groups of *P. laura* were 2.28 and 3.27 times higher than in the control group, respectively.

3.2. The effect of zinc on the growth, development, and reproduction of *D. melanogaster*

There were significant differences in egg laying amount, hatching, pupation and eclosion rates, and developmental duration of *D. melanogaster* under the effect of Zn concentration (Table 1; LSD test, $P < 0.05$); the hatching, pupation, and eclosion rates of *D. melanogaster* in

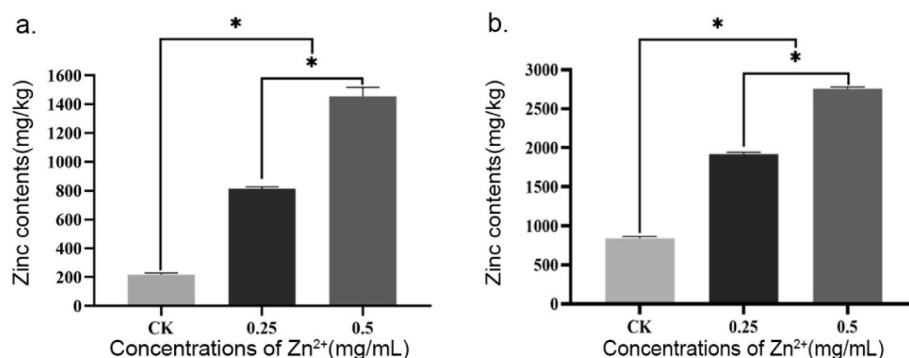


Fig. 1. Zinc contents in (a) *D. melanogaster* under different Zn²⁺ concentrations and (b) *P. laura* fed *D. melanogaster* with different Zn²⁺ concentrations. An asterisk indicates a highly significant difference between groups (LSD test, $p < 0.001$). The data are shown as mean $\pm$ SE.

Table 1

Biological properties of *D. melanogaster* under different Zn²⁺ concentrations.

Biological properties	Concentrations of Zn ²⁺ (mg/mL)		
	CK	0.25	0.50
Egg amount	110.333 $\pm$ 2.789a	88.467 $\pm$ 1.895b	81.523 $\pm$ 2.162b
Hatching rate (%)	92.274 $\pm$ 0.034a	82.042 $\pm$ 0.066b	73.174 $\pm$ 0.059c
Pupation rate (%)	83.122 $\pm$ 0.022a	69.057 $\pm$ 0.036b	66.058 $\pm$ 0.051c
Eclosion rate (%)	77.169 $\pm$ 0.016a	67.799 $\pm$ 0.018b	65.950 $\pm$ 0.023c
Developmental duration (d)	8.533 $\pm$ 0.236a	10.467 $\pm$ 0.307b	12.333 $\pm$ 0.304c

The data are shown as mean $\pm$ SE. The data with different letters denote a significant difference (LSD test, $P < 0.05$).

the 0.50 mg/mL Zn concentration group were significantly lower than those in the 0.25 mg/mL Zn concentration group and the CK group (LSD test, $P < 0.001$), and the developmental duration of *D. melanogaster* was significantly longer (LSD test, $P < 0.001$). In Fig. 2a and b, the body weights of both female and male *D. melanogaster* at Zn concentrations of 0.25 mg/mL and 0.50 mg/mL were significantly lower than those of the CK group (LSD test, $P < 0.05$). There was a significant reduction in body length of both female and male *D. melanogaster* under the three different treatments (Fig. 2c and d; LSD test, $P < 0.05$).

Moreover, there was also a significant difference in the longevity of *D. melanogaster* (log-rank test, $P < 0.05$), with the medium Zn concentrations of 0.25 mg/mL and 0.50 mg/mL having significantly shorter lifespans than the CK group (Fig. 3). In addition, according to the survival curve analysis, the survival rate of *D. melanogaster* with Zn medium concentrations of 0.25 mg/mL and 0.50 mg/mL were significantly lower than the CK group (log-rank test, $P < 0.05$).

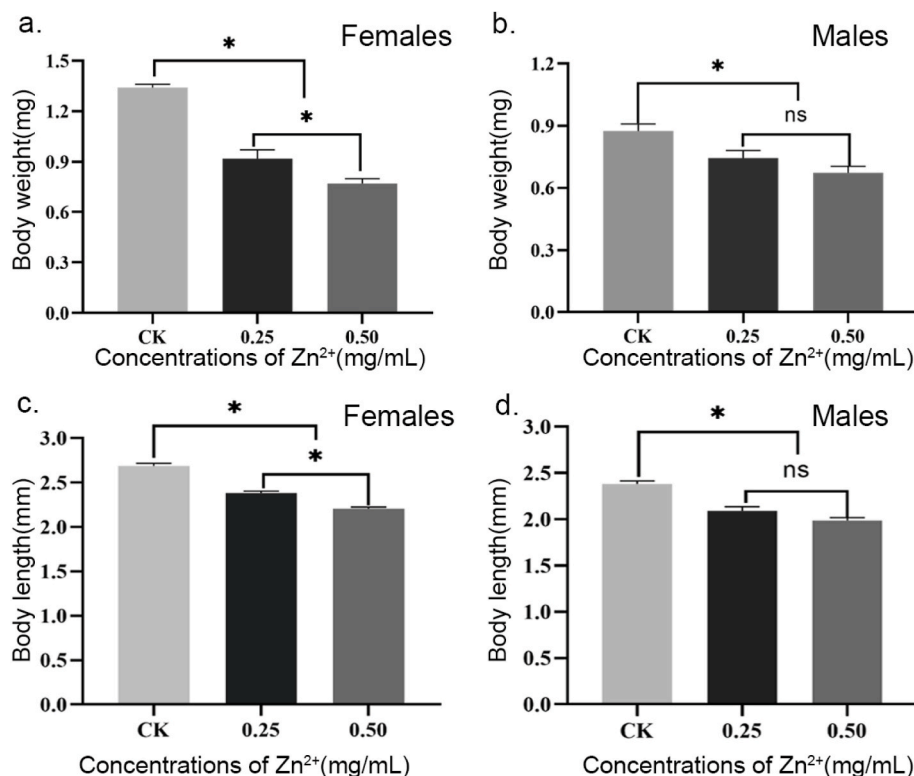


Fig. 2. Body weight (a, b) and length (c, d) of *D. melanogaster* under different Zn²⁺ concentrations. An asterisk indicates a highly significant difference between groups (LSD test, $p < 0.05$). The ns indicates no significant difference (LSD test, $p > 0.05$). The data are shown as mean $\pm$ SE.

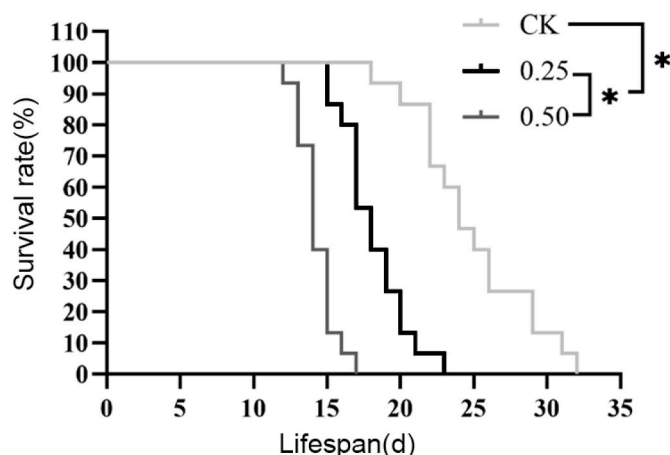


Fig. 3. Survival rate and lifespan of *Drosophila melanogaster* under different Zn^{2+} concentrations. CK, 0.25, and 0.50 denote different Zn^{2+} concentrations (mg/mL). An asterisk indicates a highly significant difference between groups (log-rank test, $p < 0.05$).

3.3. The effect of zinc on the growth, development, and reproduction of *P. laura* through the food chain

The survival rate of first-generation *P. laura* from hatching to maturity significantly decreased from 70.00% to 46.67% and 38.33% under the three different Zn^{2+} concentration treatments (chi-squared test, $P < 0.05$; Fig. 4a). Multiple group rate comparisons by the chi-squared test revealed significant differences in *P. laura* survival rates between the CK group and both the 0.25 mg/mL and 0.50 mg/mL Zn^{2+} concentration treatment groups ($P < 0.005$) but not between the 0.25 mg/mL Zn^{2+} group and the 0.50 mg/mL Zn^{2+} group ($P > 0.05$). When *P. laura* were fed *D. melanogaster* reared on different Zn concentration mediums, there were significant differences in each instar and in total developmental duration of *P. laura* (LSD test, all $P < 0.05$) (Table 2).

Body weight was significantly lower in *P. laura* with Zn concentrations of 0.25 mg/mL and 0.50 mg/mL than in the CK group (LSD test, $P < 0.05$; Fig. 4b). Body length was significantly greater in *P. laura* in the CK group than in the Zn concentrations of 0.25 mg/mL and 0.50 mg/mL (LSD test, $P < 0.05$; Fig. 4c). The egg laying rate of *P. laura* in the CK group was significantly higher than in the 0.25 mg/mL and 0.50 mg/mL Zn concentration treatment groups (chi-squared test, $P < 0.05$; Fig. 4d). As shown in Fig. 4e, the preoviposition period of *P. laura* was significantly prolonged from 3.64 days to 4.91 and 5.82 days under the three different Zn concentration treatments (LSD test, $P < 0.05$), and the preoviposition period of the Zn concentration of 0.50 mg/mL treatment group was significantly higher than that of the Zn concentration of 0.25 mg/mL treatment group and the CK group (LSD test, $P < 0.05$). In addition, there was a significant reduction in egg laying amount of *P. laura* with increasing Zn concentration (LSD test, $P < 0.05$) (Fig. 4f).

4. Discussion

4.1. Enrichment of zinc through the food chain

It is well known that the food chain is an important pathway for the transfer of toxic metals. When the level of toxic metals exceeds the self-regulation range, enrichment of toxic metals occurs in an animal's body. As spiders may account for up to 75% of the prey of small mammals and birds (Hunter et al., 1987; Naef-Daenzer et al., 2000), their role in the transfer of Zn along the food chain is likely to be important. The extent of toxic metal accumulation in spiders varies according to sex, physiological condition, and individual detoxification capacity (Hendrickx et al., 2003; Stalmach et al., 2015; Wisniewska et al., 2022), so the present study did not measure Zn metal content in the experimental

species by sex and body part but instead aimed to represent the enrichment of Zn metal at the hierarchical scale of the food chain. The goal was to show the level of enrichment of Zn at the food chain level.

We found that the toxic metal zinc could be transferred through the *P. laura* food chain and biomagnified at different trophic levels with remarkable results. Our study is consistent with that of Hendrickx et al. (2003), who found that the toxic metal cadmium can be transmitted through the food chain in such a way that its levels increase in spiders. Crawford et al. (1990) concluded that the toxic metal copper does not increase in the beet aphid through the food chain. This is because organisms adopt different detoxification strategies for toxic metals in their bodies, such as differences in absorption and excretion, thus reducing the toxic effects produced by toxic metals, while the neutralisation strategies of organisms for excess metals in their bodies are also related to the properties of the metals themselves (Babczynska et al., 2011).

The concentrations of metals in the spiders depended on the way they were absorbed and neutralised, including binding to molecules such as metallothionein (MT), or elimination through moulting or excretion (Babczynska et al., 2011). Due to their polyphagy and easy absorption of nutrients, these carnivorous invertebrates can bioaccumulate higher concentrations of toxic metals than other invertebrates (Hendrickx et al., 2003; Hunter et al., 1987). This is the reason for the significant bioconcentration that can occur in spider predatory food chains. The midgut diverticulum of spiders is significantly more efficient at assimilating metals (Hopkin et al., 1989), and Zn (Breymeyer and Odum, 1969), which is also excreted very slowly by spiders, is responsible for the significant biomagnification of spiders compared to other insect predators (Hendrickx et al., 2003).

The Zn content in *P. laura* reared on a medium with a Zn concentration of 0.50 mg/mL was 1.434 times higher than in the 0.25 mg/mL group, while in the *D. melanogaster* group, the difference in Zn content was 1.744 times, with a decreasing difference in content compared to the 2-fold difference in Zn content in the medium. It has been shown that as the concentration of exogenous toxic metals increases, the spider's ability to control the concentration becomes stronger, thus resulting in the spider's body zinc content not directly reflecting the zinc content of its food. Furthermore, as species become more capable of dealing with excess toxic metals in their bodies as they move up the food chain level, it can be inferred that the biomagnification of Zn in higher predators in the food chain does not increase significantly with increasing levels of Zn in the food source. However, in this experiment, in organisms higher up in the food chain, the level of zinc metal in the species becomes higher and higher, thus enabling us to assume that even when zinc metal contamination does not occur at very high concentrations, it can still be at a high level in the organism after enrichment through the food chain. This would perhaps be related to the response of organisms themselves to Zn, an essential micronutrient for which organisms have developed different mechanisms for uptake and utilisation (Butt et al., 2018), e.g., some Hymenoptera and Lepidoptera species excrete the metal in their faeces, thus limiting their accumulation at higher trophic levels (Lindqvist, 1992). Therefore, as trophic levels increase, it may not be possible to show highly significant differences in the levels of toxic metals in vivo within the same trophic level.

4.2. The effect of zinc on the growth, development, and reproduction of *D. melanogaster* and *P. laura* through the food chain

It was recognised in early studies that toxic metals could pose potential health risks to plants, carnivores, and humans through the food chain (Blakbern, 2003; Prince et al., 2001). The effects of toxic metal contamination on animal growth, development, and reproduction have been a widely studied issue, with biological characteristics, such as body weight and length of animals, being important indicators of good development. In the present study, we found that zinc had a significant negative effect on the growth, development, and reproduction of *D. melanogaster* and *P. laura* by means of food chain transfer. This is

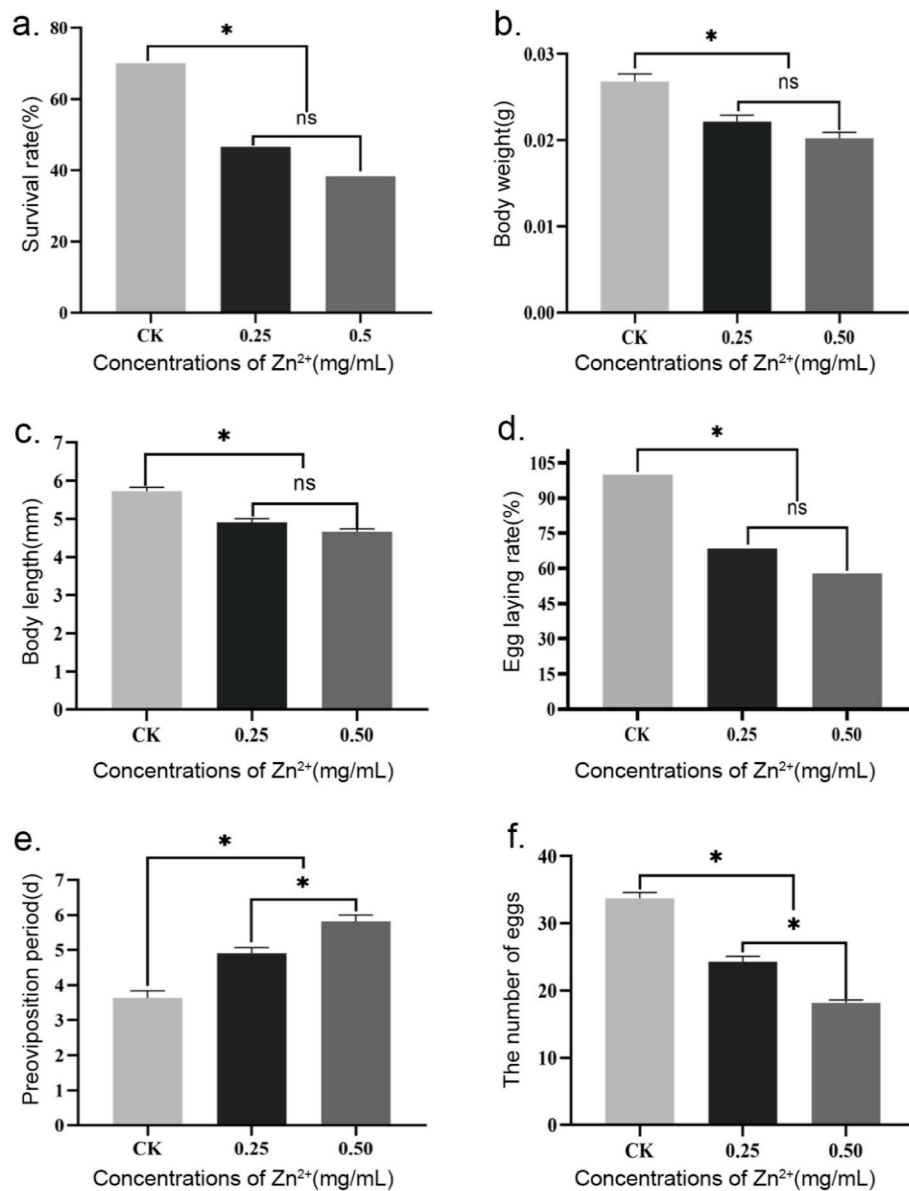


Fig. 4. (a) Survival rate, body (b) weight and (c) length, (d) egg laying rate, (e) preoviposition period, and (f) the number of eggs of *P. laura* under different Zn²⁺ concentrations. An asterisk indicates a highly significant difference between groups (LSD test, $p < 0.05$). The ns indicates no significant difference (LSD test, $p > 0.05$). The data are shown as mean $\pm$ SE.

Table 2

Developmental duration of *P. laura* under different Zn²⁺ concentrations.

Developmental stage	Concentrations of Zn ²⁺ (mg/mL)		
	CK	0.25	0.50
3rd instar	8.90 $\pm$ 0.45a	9.66 $\pm$ 0.55b	10.20 $\pm$ 0.70c
4th instar	7.52 $\pm$ 0.74a	8.70 $\pm$ 0.59b	10.63 $\pm$ 0.95c
5th instar	8.50 $\pm$ 0.82a	10.00 $\pm$ 0.74b	13.23 $\pm$ 0.92c
6th instar	9.21 $\pm$ 0.78a	12.60 $\pm$ 0.98b	15.06 $\pm$ 0.76c
7th instar	9.79 $\pm$ 0.81a	14.71 $\pm$ 0.78b	17.04 $\pm$ 0.84c
Total	43.53 $\pm$ 3.01a	54.87 $\pm$ 2.77b	64.28 $\pm$ 2.67c

The data are shown as mean $\pm$ SE. The data with different letters denote a significant difference (LSD test, $P < 0.05$).

consistent with the results of Eraly et al. (2011), which suggested that toxic metals inhibit insect growth and development, and Zhang and Zhao (2009), which showed that the developmental duration of *P. laura* was prolonged and its body weight was reduced in response to Zn. It has

also been reported that toxic metal poisoning may not affect larval survival if aquatic insects reallocate energy required for growth and development to metal detoxification processes, but this reallocation of energy can lead to the onset of developmental delays (Anderson et al., 1980; Arambourou et al., 2020; Debecker et al., 2017; Lidman et al., 2020; Perez and Noriega, 2014). The main reason for the toxicity of toxic metals is that high concentrations cause oxidative stress and induce high levels of reactive oxygen species (ROS) production in organisms. Previous studies have shown that high concentrations of ROS damage cellular and tissue structures and affect spider brain ganglia, venom glands, and silk glands (Dai et al., 2022; Wilczek et al., 2019; Yang et al., 2018a, 2018b), inhibit the growth and development of organisms, and affect many biological processes (Fu and Xi, 2020; Gobe and Crane, 2010). Excess toxic metals could also directly inhibit the growth and development of the organism through physiological processes. Spiders threatened by the toxic effects may allocate energy resources to detoxification processes combined with a behavioural reduction in intake of contaminated food, indirectly affecting growth and development by

ultimately resulting in a reduction in biological characteristics, such as body weight, length, and survival rate.

However, in contrast to the present study, some studies have found that the addition of appropriate concentrations of Zn can increase the longevity of species and enhance their fertility (Piper and Spears, 1982). For example, a study on the wheat aphid (Green and Papavizas, 1968; Subbarao et al., 1999) found that treatment with low levels of Zn induced a corresponding metabolic compensation mechanism and to some extent enhanced fertility in the species; the reason for these results being that Zn as an essential element increases the maturation and number of sperm in animals (Gilbert et al., 1996). However, the inconsistency of the results is most likely due to the chronic intake of toxic metal-rich foods in arthropods and the enrichment of toxic metal Zn in the body, which has a negative impact on their reproductive function. This would also suggest that there are differences in the reproductive potentials of different species. However, as the egg laying amount of *P. laura* was recorded only for the first egg sac, the effect of zinc on the total egg laying amount throughout its life remains to be considered.

A point of interest in the negative effects of zinc treatment on *D. melanogaster* is that the negative effects on egg laying rate in females and on body weight and length in males did not increase significantly with increasing Zn concentrations. Most insects exhibit a high reproductive capacity, which requires large amounts of energy (Cardoso-Jaime et al., 2022), and when Zn levels in *D. melanogaster* are high and continue to rise, more energy may be allocated to the processes involved in metal detoxification. Females do not use as much energy as males to detoxify themselves, thus showing a difference in body length and weight. The number of eggs laid by female fruit flies was significantly affected by increasing Zn concentrations, while egg laying rate was not significantly affected, suggesting that the reproductive strategy chosen by fruit flies in the face of high concentrations of toxic metals is to reduce the number of eggs laid to ensure egg production rate.

Most of the species in the order Arachnida also showed high reproductive capacity. The preoviposition period and egg laying of *P. laura* females under high concentrations of toxic metals decreases significantly with increasing Zn concentrations; again, egg laying rates do not have this same tendency to decrease significantly as in *D. melanogaster*. Therefore, we can assume that highly reproductive species will adopt the same reproductive strategies in the face of high concentrations of toxic metals. Babczynska et al. (2012) reported that in the actively hunting spider *Xerolycosa nemoralis*, it was found to exhibit 'fewer but more energetic eggs' in contaminated habitats. The experimental design of this study initially did not distinguish between the biological characteristics of *P. laura* by sex, which can be explained by the physiological differences between female and male spiders. Females adopt more diverse and even more energetically demanding defence mechanisms than males to more effectively protect themselves from damage caused by applied xenobiotics to preserve genetic material and maintain reproductive health (Stalmach et al., 2015; Wilczek et al., 2017), which in turn manifests itself in males being more prone to accumulating metals than females (Wilczek et al., 2005, 2008). The growth and reproduction of males are more significantly influenced by toxic metal accumulation. The stronger defensive mechanisms of females and their choice of reproductive strategy may explain the lack of significant differences in egg laying rates with increasing Zn concentrations.

Although Zn treatment had a negative effect on survival, body weight, and length of *P. laura*, there were no significant differences with increasing Zn concentrations, as shown in Fig. 4a, b, and 4c. There was no significant difference between the two experimental groups. For one, this may be a flaw in our experimental design and statistical treatment, and the above data should have been counted separately for each sex of spider. Secondly, this could also be related to the energy allocation of the spiders during their growth and developmental stages. In the experiments of Babczynska et al. (2011), the effective concentration of protein used for detoxification in spiderlings from polluted areas was not

only lower than that of the females from each experimental group, but they were not stimulated by the metals in their food. The third may be related to the degree of toxic metal enrichment in the spiders; the previous results showed that the disparity in toxic metal content in the spiders was reduced compared to the disparity in the content in the food, but there was still a significant difference in the content within the experimental groups, which leads to the assumption that the detoxification mechanism of *P. laura* for Zn increases with its concentration. Babczynska et al. (2011) found that the percentage of MT positive cells and MT concentration were positively correlated with the Cu and Cd content of the spider's abdomen. Furthermore, the findings of Wiśniewska et al. (2023) suggest that spiders may only be exposed to sublethal concentrations of metals when they show toxic excitatory effects, enhancing the expression of proteins that may have protective effects.

Previous studies (Li et al., 2016b; Stalmach et al., 2015; Wilczek et al., 2019) have demonstrated that Cd and Cu cause oxidative damage at the cellular and molecular level in tarantulas, which in turn affects gene expression and physiological function (Dai et al., 2022), and it is the first time that many of these conclusions have been corroborated by our studies on zinc. Zinc, as an essential metal, affects organisms at the genetic level differently from other toxic metals, as shown by recent evidence on honeybees, where dietary levels of iron and zinc alter the expression of major royal jelly proteins, with higher zinc levels decreasing the expression of major royal jelly proteins associated with reproduction while increasing the expression of royal jelly proteins with defence roles (Cardoso-Jaime et al., 2022; de Almeida Longuini et al., 2021). We look forward to resolving the mechanisms of its effects and, more importantly, the mechanisms of detoxification in future studies, and the results of this study can provide guidance for subsequent studies on its effects on wolf spiders in molecular damage affecting gene expression, among others. One drawback is that this study only examined one generation of *P. laura*, whereas Peterson et al. (2017) showed that the persistence of contaminants in the environment can lead to altered mating preferences, which may lead to reduced health and long-term demographic and multigenerational effects, and that the successional effects of the prolonged presence of zinc in the environment on *P. laura* need to be further verified. The long-term presence of zinc in the environment and its successional effects on *P. laura* need further investigation.

5. Conclusion

In this study, using different Zn concentrations as factors, we found that excessive Zn exposure from food sources enriched into the spider *P. laura* through its food *D. melanogaster* and had significant negative effects on the growth and reproduction of *D. melanogaster* and *P. laura*. The developmental duration and preoviposition period of *P. laura* was significantly prolonged. Moreover, its survival rate, body size, and egg laying amount was significantly reduced. The effects of toxic metals on spiders are still not well studied and need to be further explored to improve the relationship between the cascading effects of toxic metals on ecosystems through the food chain.

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CRediT authorship contribution statement

Chong Zhan: Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Yueyi Zhang:** Conceptualization, Methodology, Data curation, Writing – original draft. **Wei Li:** Data curation, Formal analysis. **Shichang Zhang:** Methodology, Data curation. **Jie Liu:** Formal analysis. **Yao Zhao:** Supervision, Writing – review & editing. **Yu Peng:** Conceptualization, Supervision, Funding acquisition, Project administration, All authors contributed to the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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